Multiple linear regression

1338 rows x 5 columns

Two columns contains string value, so after converting using dummies need to preprocess the column to convert to digit.

Support vector machine:

Hyper param	Linear	RBF	Poly	Sigmoid
C10	0.462	-0.032	0.038	-0.039
C100	0.628	0.320	0.167	0.527
C1000	0.764	0.810	0.856	0.287
C2000	0.744	0.854	0.869	-0.593
C3000	0.741	0.866	0.859	-2.124

Decision Tree:

Criterion	Max_feature	Splitter	R2 value
Squared_error	sqrt	random	0.672
Squared_error	sqrt	best	0.734
Squared_error	log2	random	0.699
Squared_error	Log2	best	0.759
Absolute_error	sqrt	random	0.761
Absolute_error	sqrt	best	0.792
Absolute_error	log2	random	0.811
Absolute_error	Log2	best	0.822
poission	sqrt	random	0.722
poission	sqrt	best	0.643
poission	log2	random	0.581
poission	Log2	best	0.674

Random Forest:

Criterion	n_estimator	R2 value
Squared_error	10	0.801
Squared_error	100	0.823
Squared_error	1000	0.844
Absolute_error	10	0.811
Absolute_error	100	0.837
Absolute_error	1000	0.854
poission	10	0.768
poission	100	0.788
poission	1000	0.803

From the above three algorithm SVM with hyper parameter 3000 + rbf having good score, hence selecting this as a good model for this example.

Internal Use